

Exercise 1 — Exploratory Data Analysis of MILK10k

Computer Vision & Speech Recognition · Joshua Cole · 14 September 2026

1. Public GitHub repository

<https://github.com/jcole-jpg/Computer-Vision-Speech-Recognition>

EDA code: notebooks/01_eda_milk10k.ipynb (executed) · helpers: src/milk10k.py · figures: reports/figures/

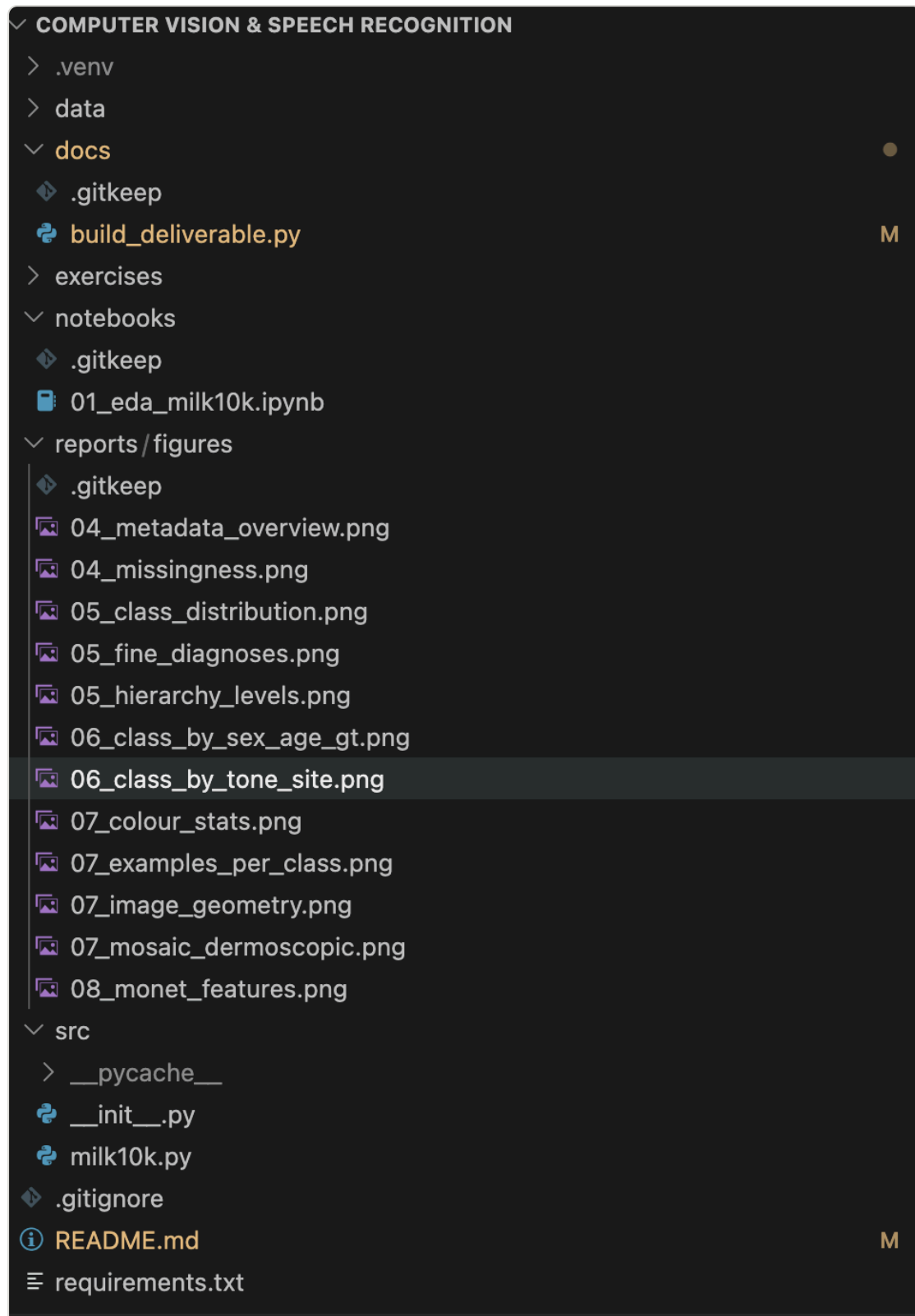
2. Tasks covered

Task	Where
Complete the EDA of MILK10k: images, metadata, class distribution, visualisations	notebooks/01_eda_milk10k.ipynb · 12 figures in reports/figures/
Choose an IDE / code editor	VS Code (screenshot, section 4)
Create a GitHub repository for course work and project	Public repo linked above
Create a project structure and push the EDA exercises	Layout on page 3; all code pushed to main
Small explanation of the clinical task	Section 3 below and README.md §1

3. Clinical task

Classify a skin lesion into its diagnostic category from a **paired clinical close-up and dermatoscopic image** plus patient metadata (age, sex, site, skin tone), so that malignant lesions (melanoma, basal cell carcinoma, squamous cell carcinoma, ...) are flagged for biopsy and benign ones are not over-treated. MILK10k provides 5,240 lesions / 10,480 images with 11 top-level classes (48 fine diagnoses); ~96 % of labels are histopathology-confirmed.

4. Project structure (code editor: VS Code)



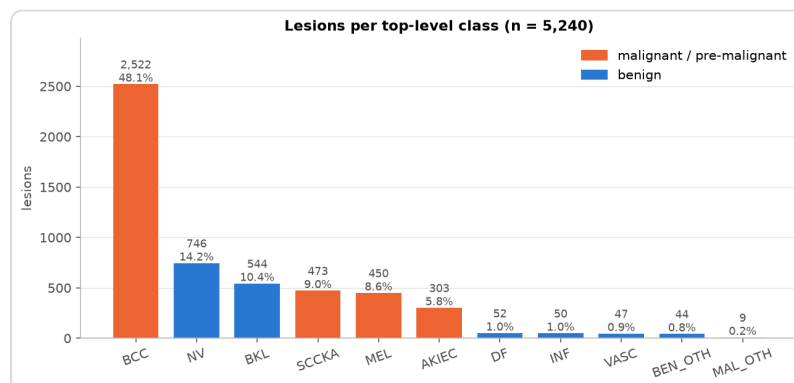
Screenshot of the project Explorer in VS Code.

5. Directory layout

Computer-Vision-Speech-Recognition/	
└─ README.md	project overview, clinical task, dataset, findings
└─ requirements.txt	Python dependencies
└─ .gitignore	keeps data / venv / checkpoints out of git
└─ data/	
└─ raw/milk10k/	extracted dataset (git-ignored)
└─ images/	10,480 x ISIC_*.jpg
└─ metadata.csv	
└─ supplements/	training_input / training_gt / training_supp .csv
└─ processed/	derived tables (git-ignored)
└─ notebooks/	
└─ 01_eda_milk10k.ipynb	Exercise 1 EDA (executed, outputs included)
└─ src/	
└─ __init__.py	
└─ milk10k.py	loaders, class map, merged image / lesion tables
└─ reports/	
└─ figures/	12 exported EDA figures
└─ exercises/	
└─ exercise_1.pdf	this deliverable
└─ docs/	
└─ build_deliverable.py	builds exercises/exercise_1.pdf
└─ editor_screenshot.png	

6. EDA summary

Finding	Implication for modelling
Exactly 1 clinical + 1 dermatoscopic image per lesion	Split by <code>lesion_id</code> ; model the pair jointly
72 % of lesions malignant; BCC alone 48 % , nevi only 14 % — biopsy-enriched cohort	Use macro-F1 / balanced accuracy / per-class sensitivity, not accuracy
Five classes with < 55 lesions (MAL_OTH has 9)	Stratified splits, class weighting / re-sampling
48 fine diagnoses, most with < 10 lesions	Train at the 11-class level
Median age 65, 60 % male, 61 % skin-tone 3, < 1 % tones 0–1	Report metrics per tone / age band
Age, site and skin tone vary systematically by class	Legitimate priors but possible shortcuts — audit
Mixed resolutions; dermatoscopy brighter, with gel / hair / ink artefacts	Per-modality normalisation and augmentation



Lesions per top-level class (n = 5,240), coloured by malignancy. Full analysis in the notebook.